MNN is a blazing fast, lightweight deep learning framework, battle-tested by business-critical use cases in Alibaba. Full multimodal LLM Android App:[MNN-LLM-Android](./apps/Android/MnnLlmChat/README.md)

www.mnn.zone/

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wangzhaode Merge pull request #3350 from xhzheng1895/kai_update

513bf36 · last week

.github	[CI:Refactor] close push/pr of intern...	2 months ago
3rd_party	MNN:Sync: Sync Interal 3.0.2	4 months ago
apps	[update] move all chat view to Views	2 weeks ago
backupcode	MNN:Sync: Sync Interal 3.0.2	4 months ago
benchmark	MNN:Sync Sync Internal 2.9.0	11 months ago
ciscripts	Github release 1.1.0	5 years ago
cmake	[MNN:Sync] Sync internal git	5 years ago
codegen	MNN:Sync: Sync Internal 2.9.5	7 months ago
demo	[MNN:Sync] Sync Internal 3.1.0.	2 months ago
doc	Doc:Bugfix: Update dingding	5 months ago
docs	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
express	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
include/MNN	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
package_scripts	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
project	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
pymnn	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
resource	Update flt_input.txt	2 years ago
schema	MNN:Sync: Sync Internal 3.1.1	last month
source	Merge pull request #3350 from xhzh...	last week
test	MNN:Sync: Sync Internal 3.1.1	last month

tools	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
transformers	LLM:Sync: Sync llmexport files	2 weeks ago
.gitignore	[MNN:Sync] Sync Internal 3.1.0.	2 months ago
.readthedocs.yaml	MNN:Docs: Fix .readthedocs.yaml, a...	11 months ago
.travis.yml	Updated travis ci build matrix. Added...	5 years ago
CMakeLists.txt	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
CONTRIBUTING.md	added CONTRIBUTING.md	2 years ago
MNN.podspec	[MNN:Sync] Sync Internal Gitlab: 2.5.1	2 years ago
MNN.sln	MNN:Sync: Sync Internal 2.9.6	6 months ago
MNN_Render.podspec	[MNN:Sync] Sync Internal 2.8.1	2 years ago
README.md	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
README_CN.md	MNN:Sync: Sync Internal 3.1.2	2 weeks ago
README_JP.md	MNN:Sync: Sync Internal 3.0.5	2 months ago
docker_release.sh	[MNN:Sync] Sync Internal 2.8.2	last year
docker_run.sh	[MNN:Sync] Sync internal Gitlab	4 years ago
release.sh	[MNN:Sync] Sync Internal 2.8.2	last year
test.bat	[Sync] Sync internal Gitlab	3 years ago
test.ps1	[MNN:Sync] Sync Internal 2.7.1	2 years ago
test.sh	MNN:Sync: Sync Internal 3.0.4	3 months ago

MNN

[中文版本]

[日本語バージョン]

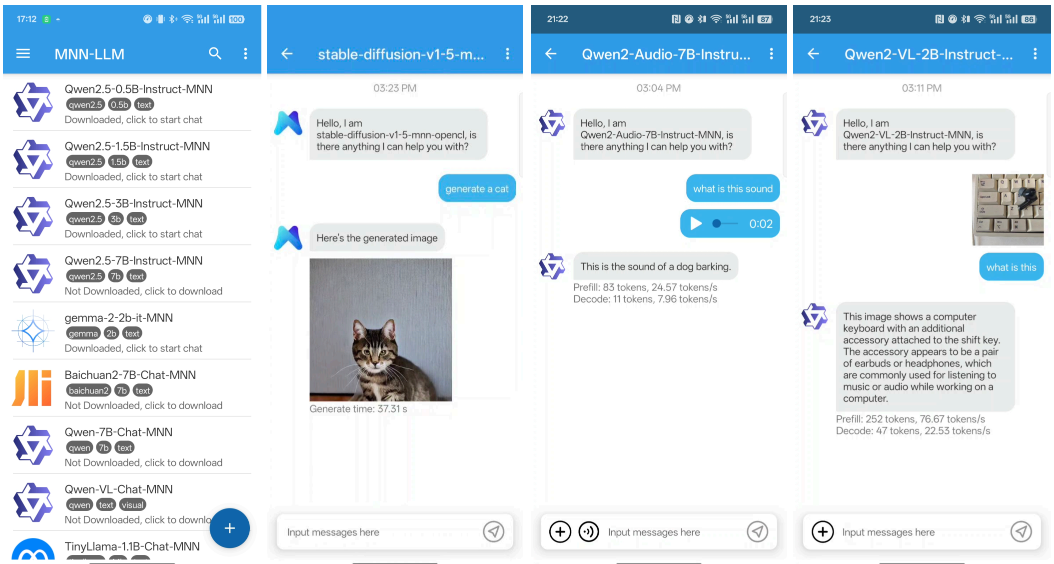
[MNN Homepage]

News 🔥

- [2025/02/18] iOS multimodal LLM App is released MNN LLM iOS.

- [2025/02/11] android app support for deepseek r1 1.5b.

- [2025/01/23] We released our full multimodal LLM Android App:MNN-LLM-Android. including text-to-text, image-to-text, audio-to-text, and text-to-image generation.

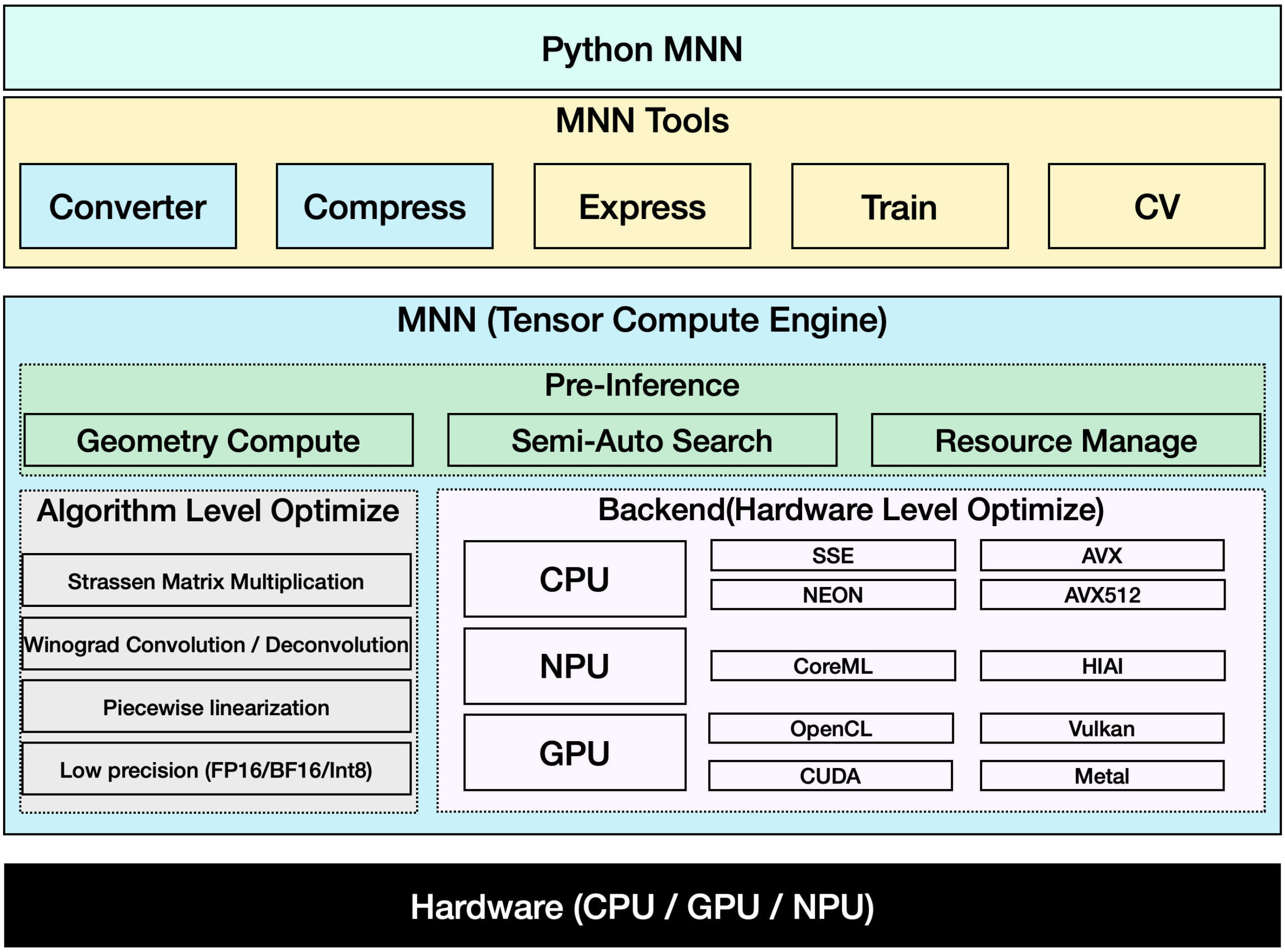


Intro

MNN is a highly efficient and lightweight deep learning framework. It supports inference and training of deep learning models and has industry-leading performance for inference and training on-device. At present, MNN has been integrated into more than 30 apps of Alibaba Inc, such as Taobao, Tmall, Youku, DingTalk, Xianyu, etc., covering more than 70 usage scenarios such as live broadcast, short video capture, search recommendation, product searching by image, interactive marketing, equity distribution, security risk control. In addition, MNN is also used on embedded devices, such as IoT.

MNN-LLM is a large language model runtime solution developed based on the MNN engine. The mission of this project is to deploy LLM models locally on everyone's platforms(Mobile Phone/PC/IOT). It supports popular large language models such as Qianwen, Baichuan, Zhipu, LLAMA, and others. MNN-LLM User guide

MNN-Diffusion is a stable diffusion model runtime solution developed based on the MNN engine. The mission of this project is to deploy stable diffusion models locally on everyone's platforms. MNN-Diffusion User guide



Inside Alibaba, MNN works as the basic module of the compute container in the Walle System, the first end-to-end, general-purpose, and large-scale production system for device-cloud collaborative machine learning, which has been published in the top system conference OSDI'22. The key design principles of MNN and the extensive benchmark testing results (vs. TensorFlow, TensorFlow Lite, PyTorch, PyTorch Mobile, TVM) can be found in the OSDI paper. The scripts and instructions for benchmark testing are put in the path "/benchmark". If MNN or the design of Walle helps your research or production use, please cite our OSDI paper as follows:

```
@inproceedings {proc:osdi22:walle,
    author = {Chengfei Lv and Chaoyue Niu and Renjie Gu and Xiaotang Jiang and Zhaode Wang
and Bin Liu and Ziqi Wu and Qiulin Yao and Congyu Huang and Panos Huang and Tao Huang and
Hui Shu and Jinde Song and Bin Zou and Peng Lan and Guohuan Xu and Fei Wu and Shaojie Tang
and Fan Wu and Guihai Chen},
    title = {Walle: An {End-to-End}, {General-Purpose}, and {Large-Scale} Production
System for {Device-Cloud} Collaborative Machine Learning},
    booktitle = {16th USENIX Symposium on Operating Systems Design and Implementation
```

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(OSDI 22)},
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    pages = {249--265},
    url = {https://www.usenix.org/conference/osdi22/presentation/lv},
    publisher = {USENIX Association},
    month = jul,
}
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Documentation and Workbench

MNN's docs are in place in [Read the docs].

You can also read docs/README to build docs's html.

MNN Workbench could be downloaded from [MNN's homepage], which provides pretrained models, visualized training tools, and one-click deployment of models to devices.

Key Features

Lightweight

- Optimized for devices, no dependencies, can be easily deployed to mobile devices and a variety of embedded devices.
- iOS platform: static library size will full option for armv7+arm64 platforms is about 12MB, size increase of linked executables is about 2M.
- Android platform: core so size is about 800KB (armv7a - c++_shared).
- Using MNN_BUILD_MINI can reduce package size by about 25%, with a limit of fixed model input size
- Support FP16 / Int8 quantize, can reduce model size 50%-70%

Versatility

- Supports `Tensorflow`, `Caffe`, `ONNX`, `Torchscripts` and supports common neural networks such as `CNN`, `RNN`, `GAN`, `Transformer`.
- Supports AI model with multi-inputs or multi-outputs, every kind of dimension format, dynamic inputs, controlflow.
- MNN supports approximate full OPs used for the AI Model. The converter supports 178 `Tensorflow` OPs, 52 `Caffe` OPs, 163 `Torchscripts` OPs, 158 `ONNX` OPs.
- Supports iOS 8.0+, Android 4.3+, and embedded devices with POSIX interface.
- Supports hybrid computing on multiple devices. Currently supports CPU and GPU.

High performance

- Implements core computing with lots of optimized assembly code to make full use of the ARM / x64 CPU.
- Use Metal / OpenCL / Vulkan to support GPU inference on mobile.
- Use CUDA and tensorcore to support NVIDIA GPU for better performance
- Convolution and transposition convolution algorithms are efficient and stable. The Winograd convolution algorithm is widely used to better symmetric convolutions such as 3x3,4x4,5x5,6x6,7x7.
- Twice speed increase for the new architecture ARM v8.2 with FP16 half-precision calculation support. 2.5 faster to use sdot for ARM v8.2 and VNNI.

Ease of use

- Support use MNN's OP to do numerical calculating like numpy.

- Support lightweight image process module like OpenCV, which is only 100k.
- Support build model and train it on PC / mobile.
- MNN Python API helps ML engineers to easily use MNN to infer, train, and process images, without dipping their toes in C++ code.

The Architecture / Precision MNN supported is shown below:

- S ： Support and work well, deeply optimized, recommend to use
- A ： Support and work well, can use
- B ： Support but has bug or not optimized, no recommend to use
- C ： Not Support

Architecture / Precision		Normal	FP16	BF16	Int8
CPU	Native	B	C	B	B
	x86/x64-SSE4.1	A	B	B	A
	x86/x64-AVX2	S	B	B	A
	x86/x64-AVX512	S	B	B	S
	ARMv7a	S	S (ARMv8.2)	S	S
	ARMv8	S	S (ARMv8.2)	S(ARMv8.6)	S
GPU	OpenCL	A	S	C	S
	Vulkan	A	A	C	A
	Metal	A	S	C	S
	CUDA	A	S	C	A
NPU	CoreML	A	C	C	C
	HIAI	A	C	C	C
	NNAPI	B	B	C	B

Tools

Base on MNN (Tensor compute engine), we provided a series of tools for inference, train and general computation.

- MNN-Converter: Convert other models to MNN models for inference, such as Tensorflow(lite), Caffe, ONNX, Torchscripts. And do graph optimization to reduce computation.
- MNN-Compress: Compress model to reduce size and increase performance / speed
- MNN-Express: Support model with controlflow, use MNN's OP to do general-purpose computing.
- MNN-CV: An OpenCV-like library, but based on MNN and then much more lightweight.
- MNN-Train: Support train MNN model.

How to Discuss and Get Help From the MNN Community

The group discussions are predominantly Chinese. But we welcome and will help English speakers.

Dingtalk discussion groups:

Group #1 (Full): 23329087

Group #2 (Full): 23350225

Group #3: QR code:

MNN开源交流/三群 in dept

This group belongs to "阿里巴巴开源" Enterprise Department Group and is only open to internal members. External people need to apply to join the organization first to access the sharing.

This QR code is valid for 365 days (before 2025-11 -27)

Historical Paper

The preliminary version of MNN, as mobile inference engine and with the focus on manual optimization, has also been published in MLSys 2020. Please cite the paper, if MNN previously helped your research:

```
@inproceedings{alibaba2020mnn,
  author = {Jiang, Xiaotang and Wang, Huan and Chen, Yiliu and Wu, Ziqi and Wang, Lichuan
and Zou, Bin and Yang, Yafeng and Cui, Zongyang and Cai, Yu and Yu, Tianhang and Lv,
Chengfei and Wu, Zhihua},
  title = {MNN: A Universal and Efficient Inference Engine},
  booktitle = {MLSys},
  year = {2020}
}
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License

Apache 2.0

Acknowledgement

MNN participants: Taobao Technology Department, Search Engineering Team, DAMO Team, Youku and other Alibaba Group employees.

MNN refers to the following projects:

- Caffe
- flatbuffer
- gemmlowp
- Google Vulkan demo
- Halide
- Mace
- ONNX
- protobuffer
- skia
- Tensorflow
- ncnn
- paddle-mobile
- stb
- rapidjson
- pybind11
- pytorch
- bolt
- libyuv
- libjpeg
- opencv

Releases 53

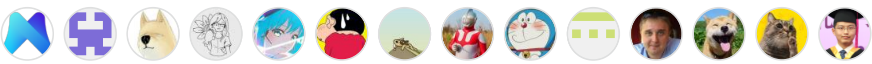
3.1.0 新增LLM移动端应用 Latest
on Feb 24

+ 52 releases

Packages

No packages published

Contributors 115



+ 101 contributors

Languages

- C++ 92.5%
- C 2.6%
- Assembly 1.5%
- Python 1.2%
- Cuda 1.0%
- Objective-C++ 0.4%
- Other 0.8%